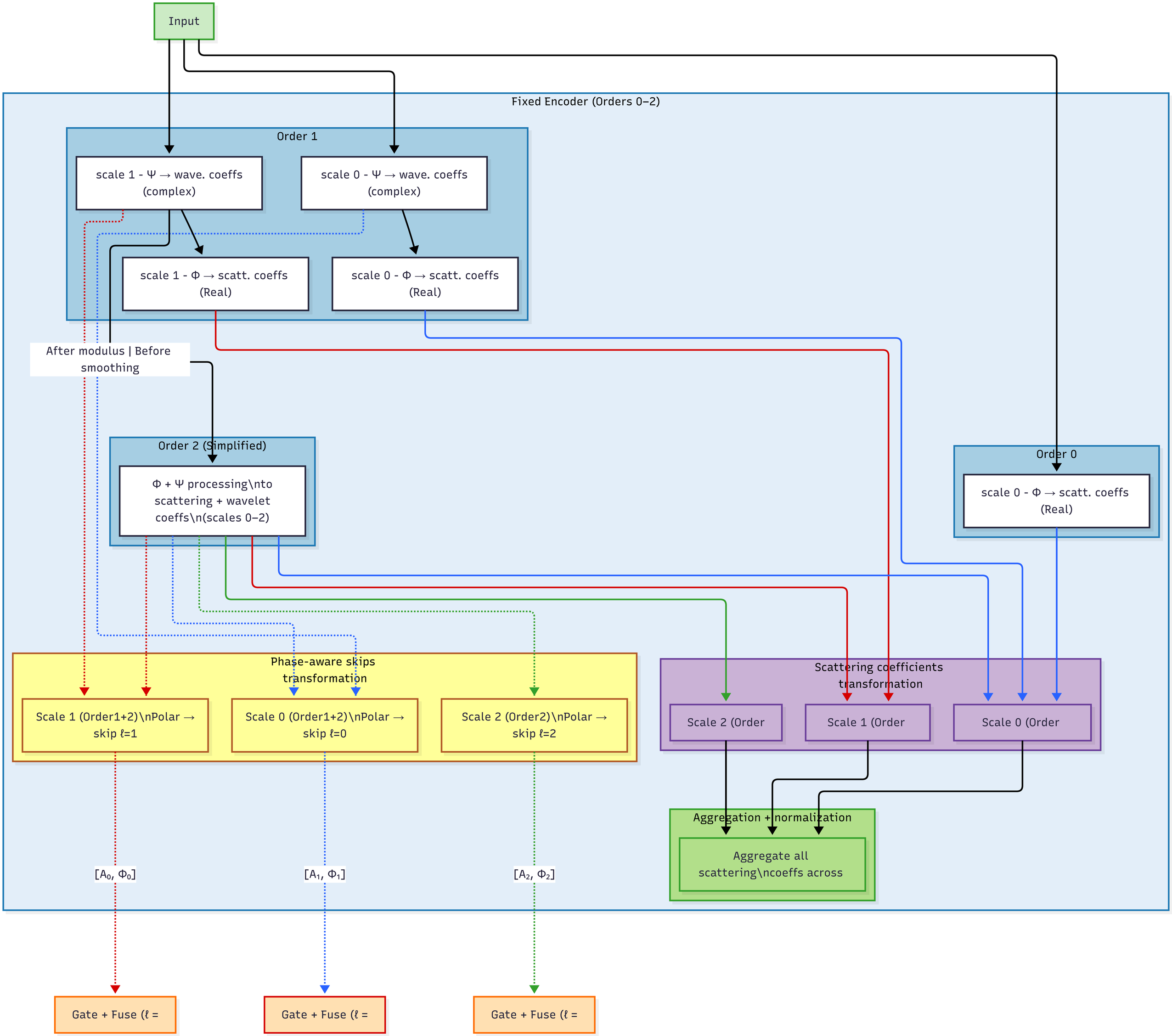
Input
Fixed Encoder (Orders 0–2)
Order 1
scale 1 - Ψ → wave. coeffs (complex)
scale 0 - Ψ → wave. coeffs (complex)
scale 1 - Φ → scatt. coeffs (Real)
scale 0 - Φ → scatt. coeffs (Real)
After modulus | Before smoothing
Order 2 (Simplified)
Φ + Ψ processing\nto scattering + wavelet coeffs\n(scales 0–2)
Order 0
scale 0 - Φ → scatt. coeffs (Real)
Phase-aware skips transformation
Scale 1 (Order1+2)\nPolar → skip ℓ=1
Scale 0 (Order1+2)\nPolar → skip ℓ=0
Scale 2 (Order2)\nPolar → skip ℓ=2
Scattering coefficients transformation
Scale 2 (Order
Scale 1 (Order
Scale 0 (Order
Aggregation + normalization
Aggregate all scattering\ncoeffs across
[A₀, Φ₀]
[A₁, Φ₁]
[A₂, Φ₂]
Gate + Fuse (ℓ =
Gate + Fuse (ℓ =
Gate + Fuse (ℓ =